THE PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
NAMANGA SECONDARY SCHOOL
FORM TWO PRE- MOCK EXAMINATION
043 MATHEMATICS

Time: 2 Hours **May 2026**

Instructions

1. This paper consists of a total of 10 questions.
2. Attempt all questions.
3. Each question carries 10 marks.
4. Mathematical tables and non-programmable calculators are not allowed
5. Write your examination number on every page of your answer sheets

1. (a) Given that $0.222\ldots$ of the school students at Seypalm secondary school are boys. What fraction of the students are girls at that school?

   (b) Three relatives, Halima, Amina, and Seypalm contributed 920 000 shillings to start a project. The ratio of contribution of Halima to Amina was 2: 5 and that of Amina to Seypalm was 3: 5. How much money did Amina contribute?

2. (a) Two pipes, A and B are used to fill a water tank. Pipe A can fill the tank in 6 hours, while pipe B can fill the same tank in 4 hours. If both pipes are opened at the same time, how long will it take to fill the tank?

   (b) If y varies directly as the square root of x, and $y = 12$ when $x = 4$, find the value of y when $x = 9$.

3. (a)It was reported that 70 000 fans attended a football game. If this number of football fans was rounded off to the nearest ten thousand what is:

   (i) The least number of fans that could have attended?

   (ii) The greatest number of fans that could have attended?

   (b) Given the following simultaneous equations $\begin{cases} 4a-3b=-1 \\ -2a+3b=2 \end{cases}$

   Find the value of x and y if $a = \frac{1}{x}$ and $b = \frac{1}{y}$.

4. (a) (i). Given that: $a^2-b^2=(a+b)(a-b)$, use it to evaluate $9\ 365 \times 9\ 365-8\ 365 \times 8\ 365$

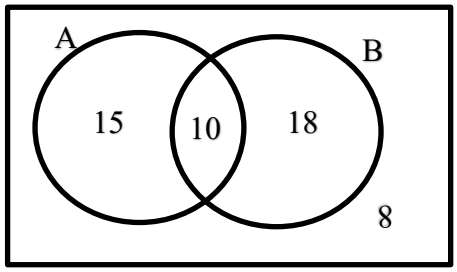
(ii). The area of a rectangle is $(16a^2\text{-}1)$ square units. Determine the possible length and width of the rectangle.

(b)Find the area of a square whose side is $(2n\text{-}5)$ units.

5. (a) When Seyplam Secondary School closes, a phone tree is used to contact each student's family. The school secretary calls three families, then each family calls other three families and so on. If the notified family on the 6th round is the same as the number of students, how many students are there in that school?

(b) Solve $\log(x^2 + 2) = \log 3x$

6. (a) Given the Venn diagram in the following figure:



How many elements are in each of the following sets?

(i) $A \cup B$

(ii) $A \cap B$

(iii) $(A \cup B)'$

(iv) $A \cap B'$

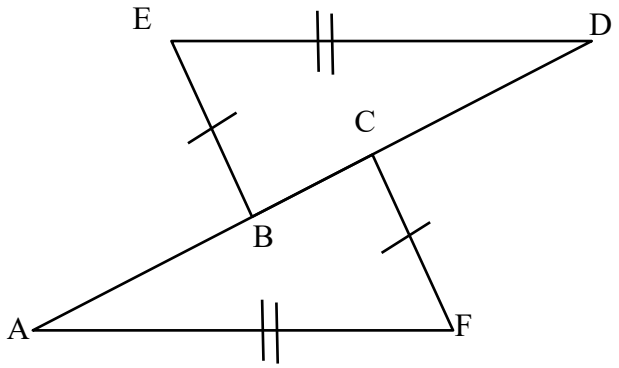
(b) Rationalize the following:

$$\frac{10}{3-\sqrt{5}} - \frac{40}{\sqrt{5}}$$

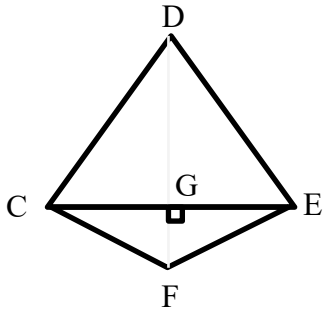
7. (a) (i). Divide 8.51 by 0.472, then give the answer correct to 6 significant figures.

(ii). Multiply 9.0017 by 0.0987 then give the answer to 2 decimal places

(b) In the following figure, $\overline{AB} = \overline{CD}$ and ABCD is a straight line. Prove that $\angle BAF = \angle CDE$



8.(a) If $\overline{DF}$ bisects ∠CDE and $\overline{CE} \perp \overline{DF}$, prove that that ΔDGC ≅ ΔDGE.



(b) In a group of 40 boys, 19 play football, 18 play volleyball, and ten play neither. How many boys play only one game?

9. (a) Without using tables or calculators, simplify the following:

$$\frac{\tan 45^o + \tan 30^o}{1 - \tan 45^o + \tan 30^o}$$

(b) An observer on top of a cliff, 25 m above sea level views a boat on the sea at angle of depression of 75°. How far is the boat from the foot of the cliff?

10. (a) Find the equation of a line passing through points (5, 2) and (t,-2) whose slope is 2

(b) Find the equation of a straight line in the form ax+by+c=0 through the point (7 ,-1) having the same x-intercept as the line 3x+5y=6.

.